

Student Management System with Admin, Teacher, and Student Roles

Rajvardhan Golande¹, Pratik Kokate², Atish Jangle³, Vaibhav Ghadge⁴, Aditya Thorat⁵, Prem Yeole⁶

1,2,3,4,5,6(Students,SRCOE, Department of Computer Engineering, Pune)

ABSTRACT: -

The Student Management System is a role-based web application designed to streamline academic operations and administrative tasks within educational institutions. This project focuses on the development of a centralized system that allows three main types of users—Admins, Teachers, and Students—to interact with the system according to their roles and privileges. Admins are provided with tools to manage users, courses, and reports, while Teachers can upload grades, mark attendance, and share learning materials. Students, on the other hand, can access their academic records, view assignments, and interact with instructors. The system is designed to replace traditional paper-based processes, thereby improving data accuracy, transparency, and accessibility. The project incorporates a secure login system, user-friendly dashboards, and dynamic content management features. By leveraging modern web technologies and databases, the system ensures efficient data flow and user management. This Student Management System aims to enhance institutional productivity, promote accountability, and foster better communication between all academic stakeholders.

Keywords: (Student Management System, Web-Based Application, Admin Portal, Teacher Dashboard, Student Portal, Academic Records, Grade Management, Attendance Tracking, Education Technology, Role-Based Access)

1. INTRODUCTION

The increasing complexity of managing academic and administrative tasks in educational institutions has created a growing need for digital solutions. Traditional paper-based systems and fragmented digital tools are often inefficient, prone to errors, and lack real-time accessibility. A centralized platform that supports role-based access can significantly enhance operational efficiency, data accuracy, and communication among stakeholders.

This research focuses on the development of a Student Management System (SMS) that incorporates three key user roles: Admin, Teacher, and Student. Each role is provided with tailored access and functionalities—Admins manage users, courses, and system settings; Teachers handle academic content, grades, and attendance; and Students access their profiles, academic records, and learning materials. The system is built using secure and scalable web technologies, aiming to improve data management, enhance transparency, and reduce the administrative workload. This paper explores the design, implementation, and practical impact of such a system, demonstrating its effectiveness in modern educational environments.

2.LITERATURE REVIEW

1.Traditional Student Information Systems (SIS): -

Early SIS implementations primarily focused on digitizing student records, attendance, and grading. These systems often lacked role-based access controls, leading to potential data security issues. Challenges included limited scalability, user-unfriendly interfaces, and inadequate support for diverse user roles.

Emerald

2. Role-Based Access Control (RBAC) in Educational Systems: -

RBAC models assign permissions to users based on their roles, enhancing security and operational efficiency. In academic settings, roles such as Admin, Teacher, and Student are defined, each with specific access rights. Implementing RBAC ensures that users can only access information pertinent to their responsibilities, reducing the risk of unauthorized data access.

3. Hierarchical and Attribute-Based Access Control Models

Hierarchical RBAC (HRBAC) introduces role hierarchies, allowing for inheritance of permissions, which simplifies management in complex organizations. Attribute-Based Access Control (ABAC) considers user attributes (e.g., department, clearance level) in access decisions, offering more granular control. Combining RBAC with ABAC can provide a flexible and scalable access control mechanism suitable for dynamic educational environments.

4.Integration of RBAC in Student Management Systems

Modern Student Management Systems incorporate RBAC to delineate functionalities among Admins, Teachers, and Students. Admins manage system configurations, user accounts, and institutional data. Teachers handle course materials, assessments, and student evaluations. Students access learning resources, submit assignments, and view their academic progress. This segregation enhances system security and ensures a streamlined user experience.

5. Security and Privacy Considerations

Implementing RBAC mitigates risks associated with unauthorized data access and potential breaches. Ensuring data privacy, especially concerning student records, is paramount and often mandated by regulations. Regular audits and updates to access control policies are necessary to adapt to organizational changes and emerging security threats.

3.FLOWCHART

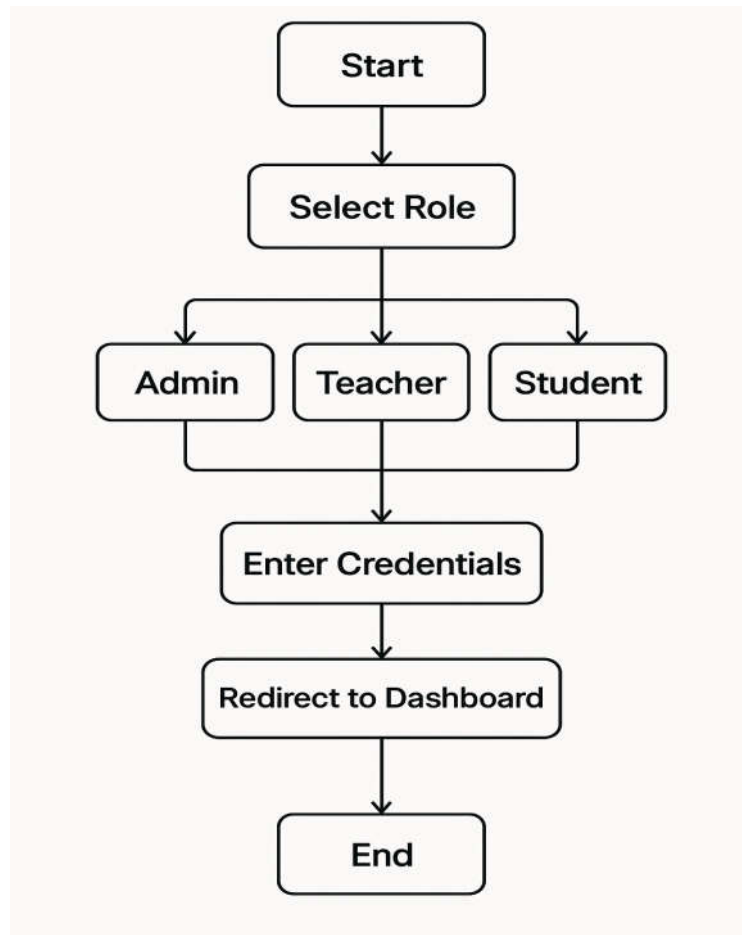


Fig no – 1 Flowchart of the system

Authentication: Ensures that users can access only the functions allowed by their roles.

Admin Role: Has control over user management and system settings.

Teacher Role: Focuses on grading, attendance, and student management.

Student Role: Allows students to check their academic progress and assignments.

The flowchart simplifies the process of understanding how each user interacts with the system and what actions they can take. It also ensures that the system is secure by restricting access based on user roles.

3.RESULT

The following image illustrates the result screen of the C++ Student Management System program:

```
===== MAIN MENU =====  
1. Admin Panel (Login once)  
2. Student Login (Enter Marks)  
3. Teacher (Generate Report)  
4. Teacher (Edit Marks)  
5. Teacher (Clear Report File and Data)  
6. Exit  
Enter choice: |
```

Fig 2- Main Menu

Fig 2. Shows the Main Menu screen of the Student Management System.

```
--- Admin Panel ---  
1. Add Student  
2. Exit Admin Panel  
Enter your choice: |
```

Fig 3- Admin Panel

Fig 3 shows the Admin Panel options to Add New Students and Exit the Admin Panel

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===== MAIN MENU =====
1. Admin Panel (Login once)
2. Student Login (Enter Marks)
3. Teacher (Generate Report)
4. Teacher (Edit Marks)
5. Teacher (Clear Report File and Data)
6. Exit
Enter choice: 2
Enter your Roll Number: 1
Enter marks for Subject 1: 50
Enter marks for Subject 2: 60
Enter marks for Subject 3: 70
Enter marks for Subject 4: 80
Enter marks for Subject 5: 90
? Marks entered successfully.

```

Fig 4 -Student login and marks entered

Fig 4 shows the Student login , student need to enter their Roll No to login and then they can Enter the marks for the the Subject

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===== MAIN MENU =====
1. Admin Panel (Login once)
2. Student Login (Enter Marks)
3. Teacher (Generate Report)
4. Teacher (Edit Marks)
5. Teacher (Clear Report File and Data)
6. Exit
Enter choice: 3
? Report generated as 'Student_Report.doc'.

```

Fig 5- Generate the report

Fig 5 shows the report file is generated . Fig 6, teacher edit the marks of the student .

```

===== MAIN MENU =====
1. Admin Panel (Login once)
2. Student Login (Enter Marks)
3. Teacher (Generate Report)
4. Teacher (Edit Marks)
5. Teacher (Clear Report File and Data)
6. Exit
Enter choice: 4
Enter Roll Number to edit: 1
Enter new marks for Subject 1: 40
Enter new marks for Subject 2: 50
Enter new marks for Subject 3: 60
Enter new marks for Subject 4: 70
Enter new marks for Subject 5: 80
? Marks updated successfully.
? Report generated as 'Student_Report.doc'.

```

Fig 6– Teacher edit marks

```

===== MAIN MENU =====
1. Admin Panel (Login once)
2. Student Login (Enter Marks)
3. Teacher (Generate Report)
4. Teacher (Edit Marks)
5. Teacher (Clear Report File and Data)
6. Exit
Enter choice: 5
??? Report file cleared and all student data deleted successfully.

```

Fig 7 – report file is cleared and all data is deleted

Fig 7 shows that using 5 Option we can clear the Report File and delete all the Data .

Analysis of C++ Student Management System Program

This C++ program implements a basic Student Management System with a console interface. It supports functionality for three types of users: Admin, Students, and Teachers. Here's a detailed breakdown of its core components and functionalities:

1.AdminPanel:-

Admin can log in using credentials (default: admin/admin123). They can add new student records, ensuring no duplicate roll numbers are entered.

2. Student Login

Students can enter their roll number. If valid and if marks haven't been entered already, students can input marks for 5 subjects.

- The data is stored in a file named 'students.txt'.

3. Teacher Functionalities

Generate Report: Creates a formatted student report in 'Student_Report.doc' with roll number, name, and marks.

Edit Marks: Allows editing a student's existing marks using their roll number.

Clear Report: Empties the report file and deletes all student data from 'students.txt'.

4. Data Management

The data is saved in a CSV format: roll number, name, sub1,sub2,sub3,sub4,sub5

File operations are used extensively to read, write, and update student information.

Overall, this program offers a functional simulation of a school system's basic operations involving students' marks handling and reporting

4. ADVANTAGES

1. Improved Efficiency: Automates time-consuming administrative tasks, reducing human error.
2. Centralized Data Management: All student information, grades, and reports are stored in one place for easy access.
3. Accessibility: Teachers, students, and administrators can access the system from anywhere with an internet connection.
4. Better Communication: Facilitates better communication between teachers and students through notifications and messages.
5. Real-Time Updates: Any changes made (e.g., grades or attendance) are reflected instantly, ensuring up-to-date information.

5.DISADVANTAGES

- 1.Dependence on Internet: Requires a reliable internet connection for access.
- 2.Security Concerns: Protecting sensitive student data can be a challenge if the system is not properly secured.
- 3.Initial Setup Cost: Developing and implementing the system might require a significant investment of time and money.
- 4.Technical Skills Required: Users (especially teachers and administrators) might need basic technical knowledge to operate the system effectively.

6.APPLICATION

- 1.Educational Institutions: The system is primarily used by schools, colleges, and universities to manage student data, grades, and communication.
- 2.Online Learning Platforms: Some e-learning platforms can integrate this system for managing virtual classrooms.
- 3.Institutional Administration: Helps in managing administrative tasks such as student enrollment, timetable management, and examination scheduling.

7. Conclusion

The Student Management System not only simplifies the complex operations of academic institutions but also enhances the quality of education by promoting organization, efficiency, and accessibility. With continuous improvements and integration of advanced technologies like AI and data analytics, such systems have the potential to revolutionize educational management in the coming years.

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